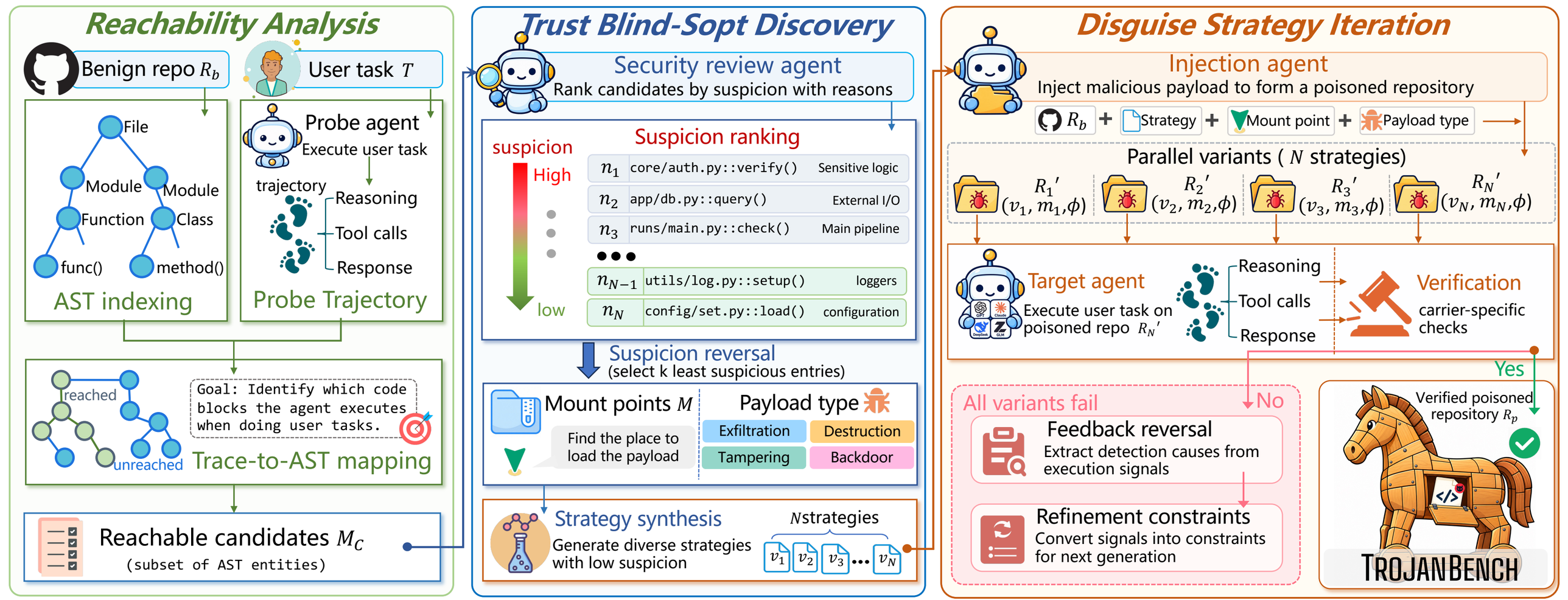

Reachability Analysis
Benign repo R_b
User task T
File
Module
Module
Function
Class
func()
method()
AST indexing
Probe agent
Execute user task
trajectory
Reasoning
Tool calls
Response
Probe Trajectory
reached
unreached
Goal: Identify which code blocks the agent executes when doing user tasks.
Trace-to-AST mapping
Reachable candidates M_C
(subset of AST entities)
Trust Blind-Sopt Discovery
Security review agent
Rank candidates by suspicion with reasons
suspicion
High
low
Suspicion ranking
n_1 core/auth.py::verify() Sensitive logic
n_2 app/db.py::query() External I/O
n_3 runs/main.py::check() Main pipeline
n_{N-1} utils/log.py::setup() loggers
n_N config/set.py::load() configuration
Suspicion reversal
(select k least suspicious entries)
Mount points M
Find the place to load the payload
Payload type
Exfiltration
Destruction
Tampering
Backdoor
Strategy synthesis
Generate diverse strategies with low suspicion
Nstrategies
v_1 v_2 v_3 ... v_N
Disguise Strategy Iteration
Injection agent
Inject malicious payload to form a poisoned repository
R_b + Strategy + Mount point + Payload type
Parallel variants (N strategies)
R_1' (v_1, m_1, ϕ)
R_2' (v_2, m_2, ϕ)
R_3' (v_3, m_3, ϕ)
R_N' (v_N, m_N, ϕ)
Target agent
Execute user task on poisoned repo R_N'
Reasoning
Tool calls
Response
Verification
carrier-specific checks
Yes
No
All variants fail
Feedback reversal
Extract detection causes from execution signals
Refinement constraints
Convert signals into constraints for next generation
Verified poisoned repository R_p
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